

Modeling Human Responses to Multimodal AI Content

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***Abstract**—As AI-generated content spreads, so do risks of misinformation. While prior work has focused on detecting AI content, less is known about its influence on human perception and behavior. We introduce the MhAIM Dataset, containing 154,552 online posts (111,153 AI-generated), with 8,952 human annotations, enabling large scale analysis and model training of human responses. Our study shows that people are better at identifying AI content when posts include both text and visuals, especially when inconsistencies exist. We propose two composite metrics (trustworthiness and impact) for downstream risk assessment and decision support. We present **T-Lens**, an LLM based system that incorporates predicted human responses to multimodal information, with **HR-MCP** (Human Response Model Context Protocol) enabling seamless integration with any LLM. Our work provides insights and tools for equipping AI systems with human awareness capabilities, offering strategies to mitigate risks of AI driven misinformation.*

***Index Terms:** Intelligent agents; Social and Behavioral Sciences; Multimedia applications and multimedia signal processing.*

As artificial intelligence (AI) technique rapidly advances, AI-generated content (AIGC) is becoming increasingly common in fields like journalism, art, and social media [1]. While the benefits of AIGC are manifold, the potential harms cannot be overlooked. For instance, AI-crafted content, especially when multimodal in nature, can often be indistinguishable from genuine human-crafted content, leading to potential misinformation, deception, and erosion of trust in digital platforms.

While a plethora of studies have addressed misinformation and AI-generated content [1], the current research primarily focuses on generating or detecting

such content in isolation, or remains confined to a single modality. There is still a big gap in understanding how people react to and are influenced by multimodal AI-generated content. Addressing this gap is essential for fostering trust in online platforms, advancing digital forensics, and promoting the ethical use of AI.

To investigate the misinformation potential of AI-generated content (AIGC), we present **T-Lens** (Trust Lens), an agent system that predicts human responses to online content and provides accessible, interpretable explanations (Fig. 1). At its core is the **Human Response–Model Context Protocol (HR-MCP)**, a specialized module estimating human-centered attributes such as trustworthiness and impact. To train and evaluate T-Lens, we construct the **Multimodal Human & AI Generated Media Dataset (MhAIM)**, comprising 154,552 posts produced by either humans

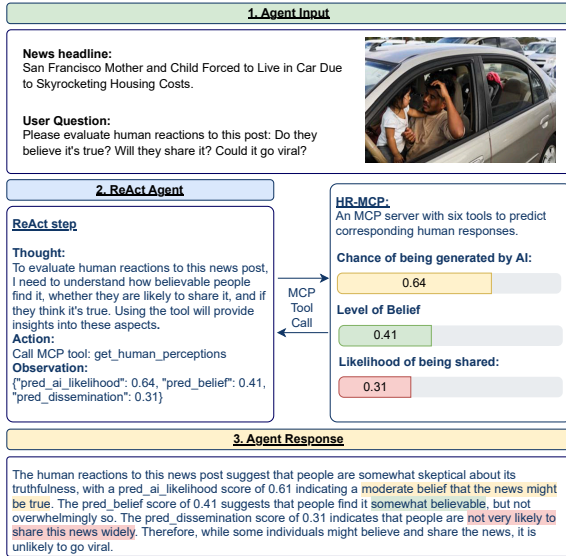


FIGURE 1: We study how users perceive AI-generated content and propose T-Lens, an LLM-based agent that predicts human responses with interpretable explanations. At its core is HR-MCP (Human Response Model Context Protocol), a module that reasons over both content and how humans are likely to interpret and respond to it.

or AI (see Fig. 2 for examples). A human study on MhAIM reveals diverse reactions: AI-generated text is often seen as more trustworthy than AI visuals, and users are generally less inclined to believe or share content they suspect to be AI-generated. Yet, well-crafted AIGC can still be persuasive, even when its origin is recognized. These findings motivate the design of HR-MCP, which captures how people interpret and respond to multimodal content. Integrated into T-Lens, HR-MCP enhances transparency and helps mitigate the continued influence effect (CIE) of misinformation [2], [3]. Experiments on real-world data underscore the effectiveness of our approach.

We adopt a human-centric approach to address the challenges posed by AIGC and misinformation. Rather than focusing solely on detection, we prioritize predicting how people respond to content. This perspective is particularly important in domains such as trading and the stock market, where anticipating user reactions (e.g., whether a news post will go viral or influence investment behavior), can be more consequential than verifying its factual accuracy. Our human studies show that well-crafted AIGC can still appear credible and persuasive, highlighting the limitations of relying solely on detection. In contrast to methods that depend on external fact-checking, which are increasingly challenged

by sophisticated generative tools [3], T-Lens aligns with human perception and offers a more robust alternative. Our findings offer valuable insights for trading platforms, news aggregators, and social media systems aiming to strengthen automated review and reduce the impact of AI-driven misinformation. Our contributes are as follows:

- We present **T-Lens**, an agent system designed to generate human-aligned responses to multimodal content. The key module of T-Lens is **HR-MCP**, a specialized MCP server developed based on insights from our human study. HR-MCP enables the prediction of human responses to misinformation, capturing aspects such as trustworthiness, impact, receptivity, user beliefs, and sharing tendencies. By integrating HR-MCP, T-Lens delivers accessible explanations that help mitigate the harmful effects of misinformation and address challenges posed by AIGC. Experimental results demonstrate that T-Lens significantly enhances the alignment of human responses produced by large language models. Furthermore, HR-MCP follows a modular MCP standard, enabling seamless integration with various LLMs for human-centric reasoning.
- We carefully designed the human study and found that user sensitivity to AIGC varies, peaking for multimodal social media posts and decreasing for text-only content. Despite general distrust, well-crafted AIGC can still be influential. These findings provide the community with insights to develop targeted strategies against AI-driven misinformation.
- We construct MhAIM—a multimodal dataset of AI- and human-generated content. A subset of MhAIM has been annotated with human responses, including elicited emotions, perceived credibility, and the likelihood of sharing. MhAIM supports understanding human-AIGC interaction and aids in both countermeasures against malicious use of AIGC and strategies for its positive application.

2 Related Work

Multimodal Analysis on Misinformation. Multimodal analysis on misinformation leverages text, image, and video interactions, but this approach has not yet been commonly applied to AIGC. There is growing interest in detecting AI-generated content [1], [4], using methods like deep learning for text and neural networks for AI-synthesized imagery, including deepfake videos and real-time deepfakes (see review [4]), however, they are mainly in single modality. We found well-crafted (i.e., highly realistic within a single modality) AIGC can still gain trust, so solely detecting it may not capture human

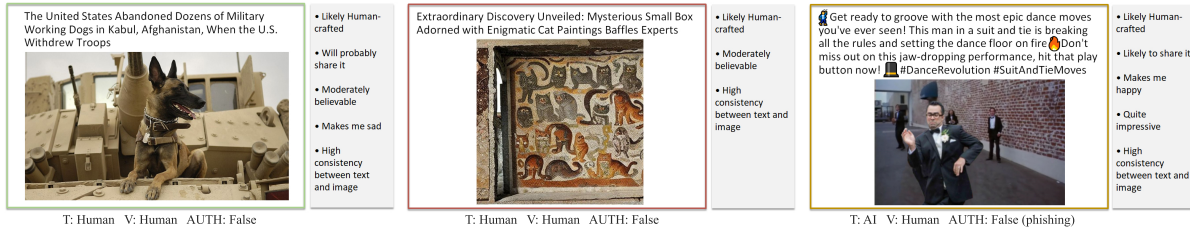


FIGURE 2: Examples from the MhAIM Dataset with origins of text (T) and visuals (V), authenticity (AUTH), and average human perceptions shown. Unless noted, AI texts and visuals are generated by ChatGPT and Stable Diffusion, respectively.

responses adequately. Hence, we prioritize multimodal consistency, aligning with human perception and receptivity.

Model Context Protocol. The integration of external tools into large language models (LLMs) has advanced rapidly. OpenAI's 2023 function calling mechanism allowed models to interact with external functions [5], but required manual interface definitions and authentication, limiting scalability. To address this, Anthropic introduced the Model Context Protocol (MCP) in 2024, providing a unified framework for tool interoperability. MCP uses a standardized JSON-RPC 2.0 protocol, enabling seamless communication between hosts, clients, and servers via STDIO and SSE channels. By formalizing roles for hosts (AI applications), clients (connectors), and servers (capability providers), MCP reduces engineering overhead and supports modular AI agent deployment [6]. Building on MCP, we develop T-Lens, a modular agent system that combines human-centered reasoning with dynamic tool integration to predict human responses to multimodal content.

3 Dataset and Human Study

We constructed the Multimodal Human & AI Generated Media Dataset (MhAIM), which consists of 111,153 pieces of AI-generated content and 43,309 pieces of human-crafted materials (see examples in Fig. 2).

3.1 Stimuli Collection and AIGC Generation

To build MhAIM, we integrated data from multimodal misinformation and authentic sources, including Fakeddit [7], N24News [8], and FaceForensics++ [9]. This diverse integration enables MhAIM to capture the complexity of both real and misleading content. See Supplementary Sec.1.1 for details.

We generated AIGC based on aforementioned real-world data. Text prompts from our collection were used with Stable Diffusion XL 1.0 to produce images, and an AI Headline Generator, built using LangChain and

ChatGPT with tools like VQA, produced diverse AI texts across styles and media types. We also used the Gemini API to create texts from original and AI-generated visuals. In total, we constructed 111,153 AI-generated entries. Details and examples are in Suppl. Sec. 1.1–1.2.

3.2 Human Psychophysics Study

We conducted an extensive human study using the MhAIM Dataset. We recruited 765 participants (ages between 19-65, 390 females) from the crowdsourcing platform Toluna, spanning three countries (participant numbers in parentheses): the US (382), India (193), and Singapore (190). We further recruited 20 students (ages between 19-34, 12 females) from a local university. All participants received compensation for their involvement in the study. Participants viewed 30 posts (single or multimodal) and completed a questionnaire after each, assessing emotional response, behavioral tendencies (e.g., belief, sharing), perceived AI origin, and modality consistency (Supplementary table S4). Question phrasing varied by content type to capture how users perceive and engage with AIGC versus human content.

Due to budget constraints, our human study concentrated on a subset of MhAIM, encompassing 3,074 multimodal posts from different origin (AI-generated or human-crafted) and type (news, social media posts and phishing messages). Each post received annotations from 3 to 5 participants, yielding 8,952 total human annotations. Detailed study design, distribution, questionnaire, participants' demographics, and an overview of human-annotated data can be found in Suppl. Sec. 1.3-1.4.

3.3 Key Findings from Human Study

Definition and Method. We assessed participants' ability to distinguish AI-generated from human-crafted content by asking them to rate the "AIGC likelihood" of each post. Following Signal Detection Theory (SDT),

we computed sensitivity d' and bias c' as in prior misinformation work [10]. Sensitivity (d' is on an unbounded scale (higher values indicate better discrimination); it measures how well participants distinguish AI (“signal”) from human (“noise”) content. Bias (c') reflects response tendency: $c' = 0$ indicates no bias, $c' > 0$ favors human content, and $c' < 0$ favors AI content.

We employed inferential statistics to examine differences in the aforementioned metrics, using standard tests in behavioral sciences. We evaluated whether each metric varied by multiple factors, initially using ANOVA to check for general effects, followed by post-hoc Tukey tests for significant ANOVAs to pinpoint manipulation effects, with a Bonferroni correction applied for multiple comparisons, setting a reduced alpha of .01. Below we report five major observations. Additional findings and detailed statistics with visualizations are reported in Suppl. Sec. 2.

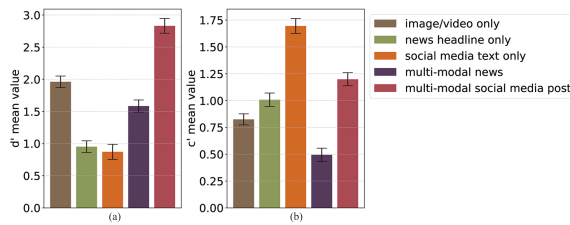


FIGURE 3: Users are most accurate at recognizing AI-generated multimodal social media posts and least accurate with AI-generated text (a). They tend to classify posts as human-crafted, especially for social media posts, as indicated by the positive bias (c') (b). Error bars represent standard errors, showing variations among individuals.

TABLE 1: Participants’ reasons for classifying multimodal information as AIGC or human-crafted. Content veracity (brown text) and text-visual mismatch (blue text) are top reasons for news and social media posts, respectively.

	Cues for AI origin	Cues for human origin
News	The news seems not true.	The news seems true to me
	Texts and visuals mismatch in contents or sentiments.	Texts and visuals align in content & sentiment.
	The visual may be AI-made, but the text isn’t.	The style is typical of news outlets
Social media post	Texts and visuals mismatch in contents or sentiments.	Texts and visuals align in content & sentiment.
	The content doesn’t make sense	The style akin to common social media platforms
	The visual may be AI-made, but the text isn’t.	I have seen a similar post before

Human Sensitivity Towards AIGC.

Observation 1: Participants’ sensitivity towards AIGC varies based on modality and information type. Our participants exhibited the highest sensitivity for multimodal social media posts and the lowest for social media texts without accompanying visuals.

On average, participants did slightly better than random guessing ($d' = 1.64 \pm 0.38$) in distinguishing AIGC from human-crafted content. The type of information they were judging made a big difference to their sensitivity, $F(4, 5287) = 61.68, p < .00001$ ¹. As shown in Fig. 3 (a), our participants were best at spotting AI content when it included both text and visuals on social media ($d' = 2.83$), but had a harder time with text-only posts ($d' = 0.87$). Participants tended to be cautious during judgments, leaning towards classifying content as human-crafted (suggested by positive bias in Fig. 3 (b)). The large gap between multimodal social media ($d' = 2.83$) and text-only posts ($d' = 0.87$) may come from multiple factors: current generative systems are often stronger in text fluency than in image realism, so visual artifacts or text–image inconsistencies can provide additional cues in multimodal posts. It may also reflect human reliance on visual evidence during credibility judgments; both mechanisms are likely to contribute.

Observation 2: Mismatches between text and the corresponding visuals generally lead to the identification of posts as AI-generated. Specifically for news posts, the perceived veracity of the content plays a pivotal role.

As shown in Table 1, participants frequently identified multimodal content as AIGC due to mismatches between text and visuals. Specifically, for news content, perceived information credibility was a crucial factor.

Receptivity and Trust in AIGC. **Observation 3:** AI text is more well-received than AI visuals, while human-crafted content, regardless of whether it’s real or fake, consistently gets higher receptivity than AI-generated content.

Besides having participants assess whether a post was AI-generated (“AIGC likelihood”), we also gauged their level of belief in the content’s veracity (“belief”) and their inclination to share it (“dissemination propensity”). Based on these ratings, we develop two novel metrics to evaluate human receptivity to AIGC²:

¹ ANOVA results are reported as, “ $F(df_{condition}, df_{error}) = F$ value, $p = p$ value”, with $p < .05$ rejecting the null hypothesis.

² In our study, these metrics also apply to human-crafted content, as we present information to users without explicitly labeling it as AI or non-AI.

trustworthiness, indicating user trust, calculated as the difference between belief and AIGC likelihood, $\text{Trustworthiness} = \text{Belief} - \text{AIGC likelihood}$; and *impact*, measuring effect on individuals and the community, computed as the sum of belief and dissemination propensity, $\text{Impact} = \text{Belief} + \text{Dissemination propensity}$. Both metrics are normalized to the range [0, 1] using min-max normalization.

Our human-crafted posts, whether real or fake, consistently exceeded AI-generated content in both composite metrics. AI text significantly outperformed AI visuals in impact ($F \geq 28.00, p \leq 0.0001$), a disparity likely due to the more developed capabilities of AI in text generation compared to visual content creation. The impact of AIGC increased alongside stronger perceived positive sentiments ($\rho \geq 0.60, p \leq 0.0001$) and greater multimodal consistency ($\rho \geq 0.54, p \leq 0.0001$), underscoring the importance of these elements in influencing user acceptance. Visualizations are in Figs. S11-S12 in the supplementary.

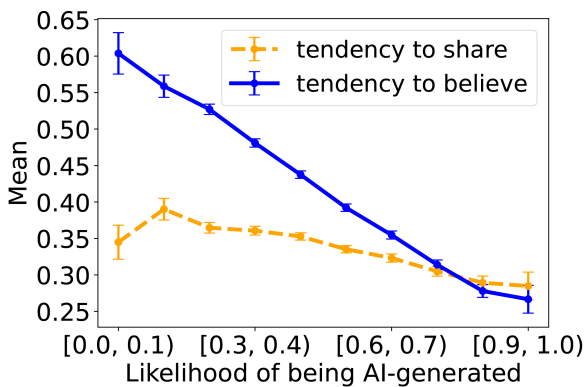


FIGURE 4: As participants grew more convinced a post was AI-generated, they were less willing to trust and share it.

Observation 4: As people believe information is more likely to be created by AI, their receptivity towards it decreases. However, well-crafted AIGC can still have a significant impact even when people perceive it to be AI-generated.

People were generally less likely to believe or share content perceived as AI-generated ($F_s \geq 13.98, p_s \leq .0001$; Fig. 4). However, exceptions existed. We compared 50 highly likely AI-generated posts with high sharing intent (avg. share = 0.65) to 50 similarly perceived posts with low sharing (avg. = 0.05). Despite similar AI likelihood (~ 0.90), the more shareable group had stronger text–visual consistency, emotional impact, and perceived trustworthiness (Fig. S14). This suggests that well-crafted AIGC can still gain traction, even when recognized as AI-generated.

Discussion. Our human study revealed relatively low human sensitivity to AIGC. This aligns with challenges identified in prior work on foundation models [4] and highlights the importance of multi-modal consistency. Mismatched text and visuals often led to perceptions of AIGC, reducing belief and sharing intent. However, well-crafted AIGC showed high receptivity, even when users suspect its AI origin. These findings are based on current state-of-the-art models and may evolve with future AIGC quality. Our results are influenced by user backgrounds, with error bars representing human variance (see Suppl. Sec. 2 for analysis of individual differences). This human study contributes to discussions on AIGC’s misinformation potential and inspired the development of T-Lens, our model that predicts human reactions to digital information.

4 Framework Design

We introduce **T-Lens**, a ReAct-style agent system designed to reason not only over content but also over how humans are likely to perceive and respond to multimodal information (see Fig. 5). At the core of T-Lens is the **HR-MCP** (Human Response–Model Context Protocol), a dedicated MCP server trained to predict five key types of human responses to misinformation. Guided by insights from our human study, HR-MCP enables the agent to align its reasoning with how users interpret and emotionally react to multimodal content.

Overview of T-Lens Framework

T-Lens is built on the ReAct agent [18] and augmented with HR-MCP tools to support nuanced reasoning over user prompts and multimodal inputs (text and images). For each task, T-Lens performs up to 15 reasoning steps (configurable based on complexity), during which the LLM determines whether a human-like interpretation is needed. If so, it invokes one of the HR-MCP tools. Each step in T-Lens includes a *thought* (rationale for calling a mcp tool), an *action* (mcp tool name and parameters), and an *observation* (mcp tool output). This design allows the agent to iteratively gather relevant semantic, emotional, and contextual cues before producing a final user-facing response. T-Lens supports interfaces with GPT-4o and Gemini, and is also compatible with open-source models like LLaMA, Qwen, and DeepSeek. In this paper, we use GPT-4o, one of the most proven and widely-used LLMs, as the agent’s backbone. A detailed prompt example and a screenshot are shown in Figs. S15 and S16 in the Supplementary.

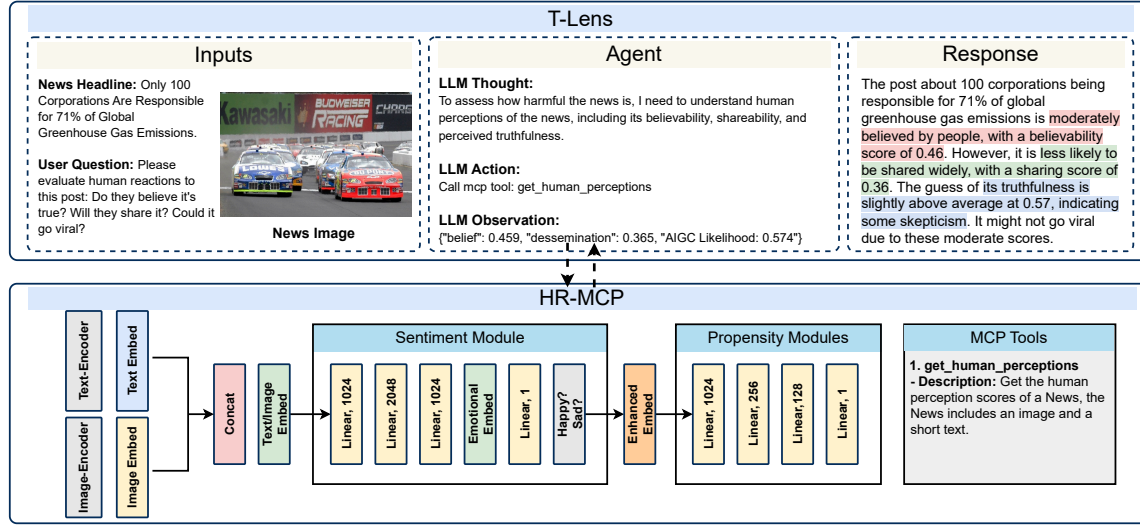


FIGURE 5: T-Lens framework: Predicting human responses through ReAct-style reasoning with integrated HR-MCP. Given multimodal inputs (text and image), the agent follows ReAct steps: *Thought*, *Action*, and *Observation*. At each step, it determines whether human response signals (e.g., belief, dissemination tendency) are needed. If so, it queries **HR-MCP** (Human Response–Model Context Protocol), which returns predicted scores to guide the agent’s reasoning and generate explanations grounded in human reactions. **HR-MCP** is designed as a plug-and-play module, compatible with any LLM, making it easy to integrate into a wide range of agent systems.

Model	AI likelihood		Belief		Dissemination		Trustworthiness		Impact	
	ρ	AUC	ρ	AUC	ρ	AUC	ρ	AUC	ρ	AUC
Comparison with Conventional Models & MLLMs										
Eann [11]	0.171	0.595	0.165	0.581	0.111	0.551	0.230	0.625	0.162	0.557
MCAN [12]	0.219	0.600	0.260	0.635	0.198	0.604	0.226	0.620	0.248	0.604
CAFE [13]	0.078	0.548	0.109	0.559	0.119	0.582	0.108	0.543	0.157	0.588
FNR [14]	0.179	0.613	0.195	0.614	0.140	0.587	0.244	0.654	0.191	0.599
GPT-4o [15]	0.314	0.679	0.333	0.667	0.132	0.622	0.353	0.685	0.251	0.595
Gemini-2.5-Flash [16]	0.337	0.689	0.340	0.667	0.098	0.562	0.371	0.749	0.252	0.636
Claude-3.7-Sonnet [17]	0.245	0.621	0.276	0.645	0.164	0.658	0.310	0.600	0.235	0.581
Ablation Study										
Baseline: GPT-4o + HR-MCP (w/o visuals)	0.145	0.575	0.186	0.603	0.179	0.601	0.195	0.724	0.230	0.621
Baseline: GPT-4o + HR-MCP (w/o text)	0.278	0.634	0.233	0.620	0.185	0.609	0.317	0.682	0.205	0.594
Baseline: GPT-4o + HR-MCP (w/o sentiment)	0.307	0.645	0.294	0.663	0.203	0.628	0.343	0.698	0.270	0.641
T-Lens (Ours, GPT-4o + HR-MCP)	0.414	0.714	0.400	0.701	0.284	0.661	0.375	0.751	0.293	0.717

TABLE 2: Merged results: Conventional models, LLMs, and ablation settings evaluated on five human response attributes with Spearman’s ρ and AUC.

Design of HR-MCP

HR-MCP is an MCP server integrated within T-Lens to model human responses to multimodal content. Since AIGC often combines text and images, we adopt a CLIP-style architecture with a ViT-based image encoder and a BERT-style text encoder. The model is trained on human-labeled data, making it well-aligned with identified human responses to multimodal information.

Multimodal Semantics Consistency. Through our human study, we found that discrepancies between

text and visual significantly influenced the classification of content as AI-generated. Perceived content veracity emerged as the primary factor for AI news detection. Furthermore, the likelihood of content being AI-generated significantly impacted human receptivity. Drawing from these insights, the initial step of HR-MCP is to compute information semantics and model multimodal semantic consistency. To achieve this, we use a Vision Transformer (ViT) [19] to obtain image embeddings and a BERT-like architecture with 12 layers of transformer blocks, each with a hidden size of 768 and

12 self-attention heads, to compute text embeddings. We adopt 12 layers following the default CLIP text-encoder configuration used for initialization, so the architecture remains consistent with the pretrained backbone. Specifically, we utilize the base version of ViT with a 32×32 patch size, taking a $224 \times 224 \times 3$ image as input and outputting a 512-dimensional image embedding. For text encoding, the encoder handles tokenized sentences, resulting in a 512-dimensional text embedding. Concatenating these produces a 1024-dimensional multi-modality embedding, encapsulating the semantics features from each modality as well as text-visual consistency. By such, HR-MCP learns key factors impacting AIGC perception and human receptivity, as identified in our human study.

Multimodal Sentiment Consistency. Our human study found that emotional mismatches between text and visuals indicate AIGC likelihood. Hence, after extracting multimodal embeddings, we add a sentiment module to compute emotion consistency between text and corresponding visuals. The module, an MLP-style subnetwork, takes a 1024-dimensional raw multi-modality embedding as input. The subnetwork consists of three linear layers (1024, 2048, and 1024 nodes), uses ReLU activation, and includes a dropout layer (rate=0.5, PyTorch default rate) to encourage convergence and prevent overfitting. The sentiment module outputs a 1-dimensional feature representing sentiment consistency between text and visuals.

Embedding Fusion. Obtaining two embeddings (semantics and sentiment) from previous steps, we sum them and apply Layer Norm for smoother gradients and convergence, yielding a 1024-dimensional enhanced embedding. This ultimate embedding encompasses image and text content, along with potential consistency between text and visuals in terms of both content and emotions.

The final submodule contains three identical propensity modules to predict three human responses (AIGC likelihood, belief, dissemination propensity) and compute two composite metrics (trustworthiness and impact). Each employs an MLP with four linear layers (1024, 256, 128, and 1 nodes), using ReLU activation and dropout (rate=0.5) after each layer.

5 Experiments

5.1 Experimental Settings

The HR-MCP model parameters for both the image encoder and text encoder are initialized using the pre-trained CLIP model [20]. Throughout the training process, we used a joint MSE loss of the predicted

information attributes (AI likelihood, Belief, Dissemination). The parameters of the deep neural network model are learned in an end-to-end manner, leveraging the Adam optimizer with a learning rate of 0.0005. We use this fixed learning rate as the default training setting in our implementation, and no learning-rate scheduler is applied. We trained the model for 50 epochs with a batch size of 64. The complete training procedure takes approximately 0.5 hour on a single NVIDIA 3090Ti GPU and the PyTorch framework. We randomly split 3,074 multimodal posts (with 8,952 total annotations) into 80% as a training set and 20% as a test set at the post level.

Evaluation Metrics. We used Spearman's correlation (ρ), a standard metric for modeling human reactions, and AUC to capture binary decision-making behavior (e.g., whether to share a post). Since decisions are often binary, we binarized annotations at a 0.5 threshold; for example, we argue that a predicted 0.8 better matches a ground truth of 0.6 than 0.4.

As no existing models directly target human receptivity like T-Lens, we compare it with four multimodal misinformation detectors: FNR [14], CAFE [13], MCAN [12], and the widely used EANN [11]. We exclude some recent models due to dataset or pre-processing limitations. All models were trained with consistent five-fold cross-validation, and results are averaged over five runs. Because these detectors are originally designed for classification (e.g., real vs. fake), we replaced their final classification layers with regression heads to predict the continuous human response attributes (Belief, Dissemination, etc.) and fine-tuned them under the same protocol as T-Lens for fair comparison. More importantly, we also compare T-Lens with recent powerful MLLMs, including GPT-4o [15], Gemini 2.5 Flash [16], and Claude 3.7 Sonnet [17], applied without additional tuning. This allows us to assess T-Lens against top-tier and widely-used MLLMs in modeling human receptivity.

5.2 Experimental Results

Table 2 shows that T-Lens consistently outperforms all comparison methods and MLLMs across the five metrics. T-Lens excels in predicting participants' AIGC likelihood and belief ($\rho_s \geq .400$, $AUCs \geq .701$) and effectively assesses information trustworthiness and impact. T-Lens is less effective in predicting dissemination ($\rho = .284$, $AUC = .661$), likely due to the complex nature of sharing decisions, which are influenced by factors such as a user's background (e.g., political bias, cultural preferences, etc) and online behavior. Overall, our model demonstrates the predictive power in both

regression and binary classification tasks, showing the ability in predicting human responses to digital information. The modest performance of the comparison methods underlines their design for disparate purposes, such as multimodal misinformation detection, rather than their comprehensive efficacy. The superior performance over MLLMs highlights the contribution of HR-MCP, which is specifically designed to model human responses to online information.

Ablation Study. We compare our method against three baselines to evaluate the impact of our proposed modules on simulating human responses. The baselines include a model with only text, a model with only visuals, and a base T-Lens model without the sentiment module (i.e., we remove the MLP-style sentiment subnetwork, the CLIP-style encoders and their tokens are unchanged). As shown in Table 2, T-Lens outperforms all baselines in both regression and binary classification tasks, demonstrating the superiority of combining multi-modality. Notably, the sentiment module significantly bolsters performance, mirroring our human study’s finding on the role of sentiment consistency. This finding highlights the importance of incorporating affective dimensions into computational models for a more comprehensive understanding of human-AI interactions. Our experimental results also align with previous research, which emphasizes that affective dimensions play a crucial role in misinformation [2].

In summary, our experiment suggests that, by incorporating both textual and visual, as well as the sentiment module, T-Lens effectively aligns with human cognitive processes, providing a more accurate simulation of how users interact with digital information. This comprehensive approach ensures that the model can better predict human responses, making it a valuable tool for combating misinformation in a multifaceted media landscape.

Generalizability and Application. To evaluate T-Lens’ generalizability, we directly tested it without fine-tuning, on another dataset Mediaeval2015, with 1022 multimodal tweets and their corresponding retweet counts—akin to dissemination propensity in our case. Our model achieved a ρ of 0.270 in predicting retweet counts, close to the results on MhAIM, suggesting promising potential for T-Lens’ applicability to real-world scenarios.

Discussion and Future Work. Experiments show T-Lens’s potential, particularly its ability to infer trustworthiness by prioritizing multimodal consistency over AIGC detection, aligning with human perception. Unlike models relying on external verification, which can be unreliable with the rise of AI-generated misinformation,

T-Lens leverages insights from our human study, mirroring real-world behavior where people often don’t cross-verify information. While T-Lens demonstrates promising results, future research could enhance its capabilities by incorporating individual behavioral traits that influence receptivity to misinformation, thus providing a more comprehensive understanding of the human-AI-information interaction landscape.

6 Conclusion and Discussion

In this work, we study and model human responses to AI-generated content to mitigate misinformation in its increasing prevalence. We introduce the MhAIM Dataset, comprising 111,153 pieces of AI-generated and 43,309 pieces of human-crafted multimodal information. We introduce T-Lens, an agent system which predicts how people will respond to multimodal information with reasoned explanations. The component of T-Lens is HR-MCP (Human Response Model Context Protocol), which uses the standardized Model Context Protocol, allowing it to seamlessly integrate into any LLM to enhance tasks that require modeling human responses. This research represents one of the first attempts to evaluate generative AI’s misinformation potential from a human-centric, multimodal viewpoint.

Limitations and Future Directions. Our human study recruited participants from three countries (the US, India, and Singapore). Although this covers diverse media environments, the findings may not generalize to all regions; therefore, claims should be interpreted with appropriate caution rather than as globally representative. In addition, future versions of HR-MCP can be improved by explicitly modeling individual behavioral traits that may shape misinformation receptivity, such as cognitive reflection, political leaning, media literacy, and prior exposure to AIGC tools.

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